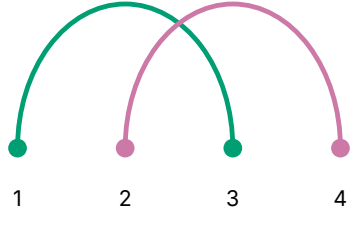
E_5: $U = \{1, 2, 3, 4\}$ has partitions 12|34 and 13|24

shorten at 5: E_4 (circuits avoiding 5);
the partition 13|24 survives, so still non-embedded



E_4 shortened at 1 is E_3:
no set has two partitions,
so E_3 is embedded

E_4 is minimal; E_5 is not